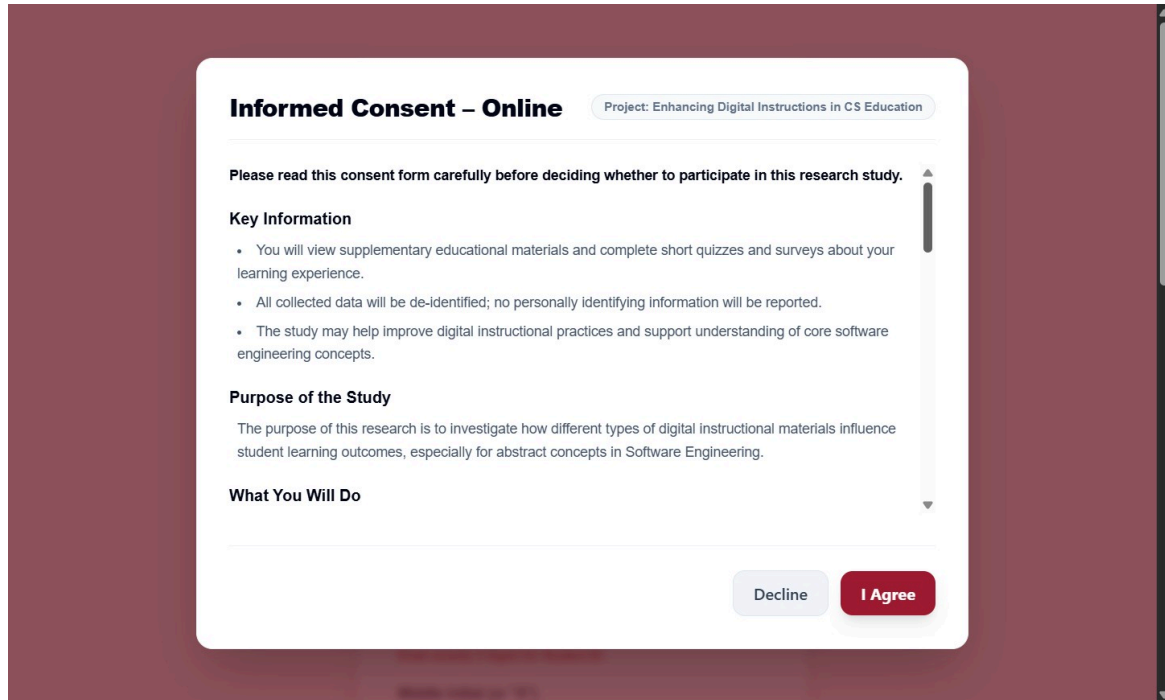


Step 1: Navigate to <https://myalp.cs.ua.edu/register>

Read the consent form



Informed Consent – Online Project: Enhancing Digital Instructions in CS Education

Please read this consent form carefully before deciding whether to participate in this research study.

Key Information

- You will view supplementary educational materials and complete short quizzes and surveys about your learning experience.
- All collected data will be de-identified; no personally identifying information will be reported.
- The study may help improve digital instructional practices and support understanding of core software engineering concepts.

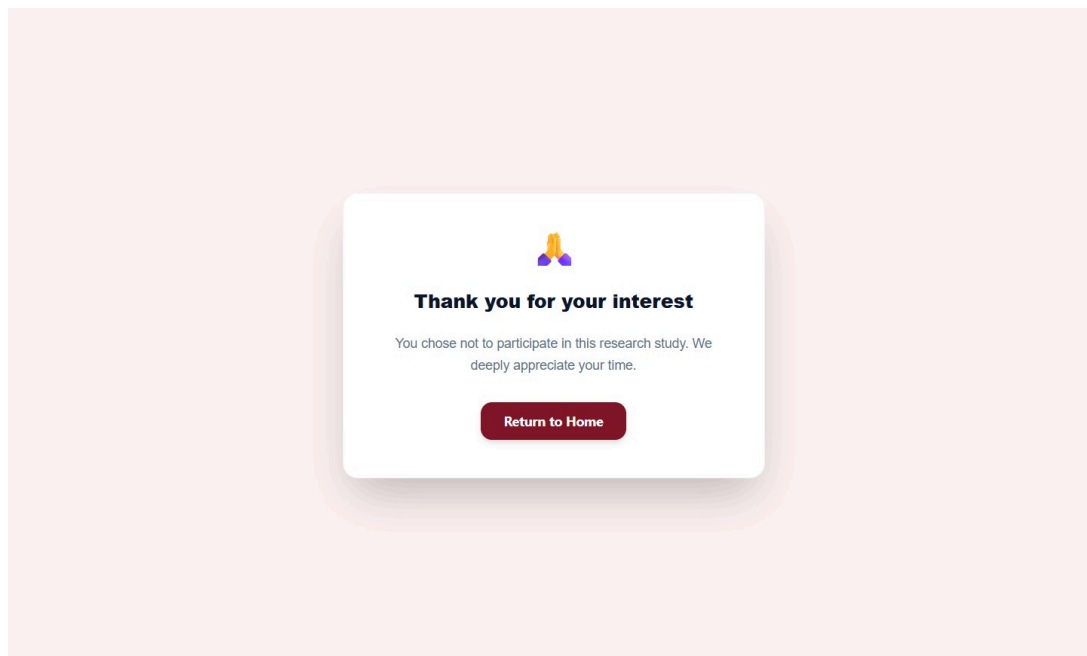
Purpose of the Study

The purpose of this research is to investigate how different types of digital instructional materials influence student learning outcomes, especially for abstract concepts in Software Engineering.

What You Will Do

Decline I Agree

A click on Decline button, thank you message is shown



Step 2: A click on “I Agree” button send participants to the sign up page for registration



Create Account

No Personal Identifiable Information required. Your userID will be automatically generated below.

 Please remember your password. It cannot be recovered if forgotten.

School Name

Auburn University

Gender

Select your gender

Gender is required.

Last 2 digits of Student ID

E.g. 45

Enter exactly 2 digits for Student ID.

Middle Initial (or "X")

Month of Birth (01-12)

E.g. 02

Enter month as two digits (01-12).

Level of Help Needed

Select your level

Select your help level.

Month of Birth (01-12)

23

Enter month as two digits (01-12).

Level of Help Needed

Building upon existing knowledge

Generated Account User ID

25-T-12

Fill required fields above...

Password

.....

Security Standard

At least 8 characters

Contains a number (0-9)

Contains a special character (e.g. !@#%\$%)

Confirm Password

.....

School Name

Auburn University

Gender

Male

Last 2 digits of Student ID

E.g. 45

Enter exactly 2 digits for Student ID.

Middle Initial (or "X")

E.g. X

Enter one letter or "X".

Month of Birth (01-12)

E.g. 02

Enter month as two digits (01-12).

Level of Help Needed

Select your level

Select your help level.

Month of Birth (01-12)

12

Level of Help Needed

Building upon existing knowledge

Generated Account User ID

25-T-12

Password

.....

Security Standard

At least 8 characters

Contains a number (0-9)

Contains a special character (e.g. !@#%\$%)

Confirm Password

.....

Robot Prevention

Generated Account User ID
25-T-12

Password
.....

Security Standard

✓ At least 8 characters

✓ Contains a number (0-9)

✓ Contains a special character (e.g. !@#%\$%)

Confirm Password
.....

Robot Prevention

What is 5 + 6?

11

refresh

Register Account

Already have an account? [Sign in](#)

Step 3: Login to access the account

Welcome Back
Sign in with your user ID

User ID

E.g. 45-X-02

Format: studentID-MiddleInitial-BirthMonth (e.g. 45-X-02)

Password

.....

Robot Prevention

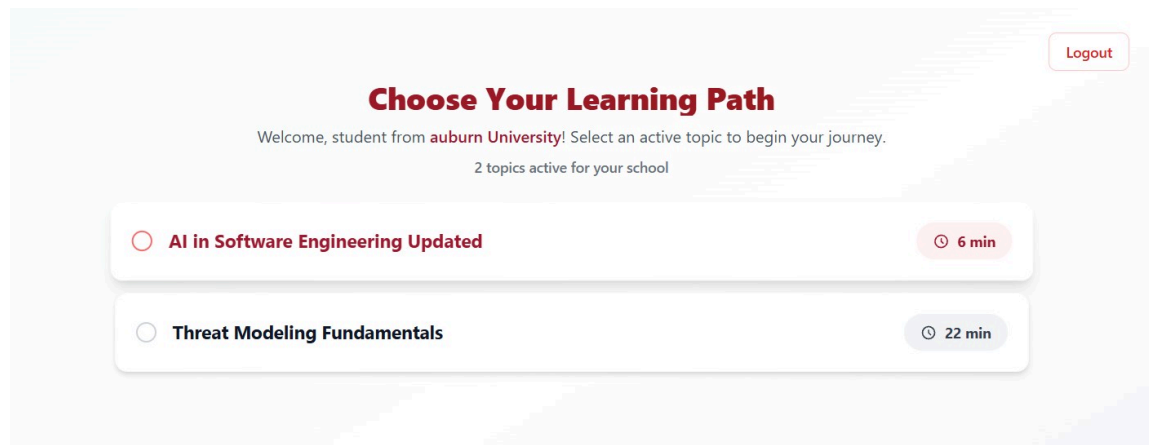
6 + 8 =

refresh

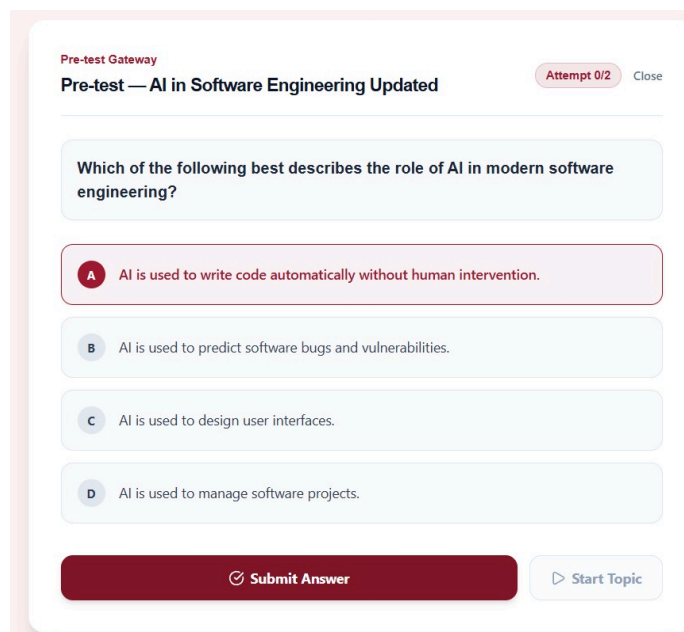
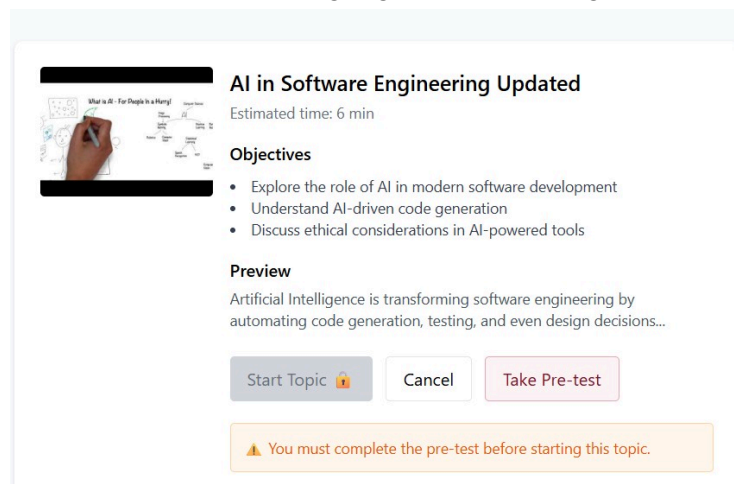
Sign In

Don't have an account? [Sign up](#)

Step 4: User select available topics for the institution



Step 5: Take Pre-Test to gauge understanding



Step 6: Start accessing learning material for the chosen Topic



AI in Software Engineering Updated

Estimated time: 6 min

Objectives

- Explore the role of AI in modern software development
- Understand AI-driven code generation
- Discuss ethical considerations in AI-powered tools

Preview

Artificial Intelligence is transforming software engineering by automating code generation, testing, and even design decisions...

Start TopicCancelRetake Pre-test

Step 7: View the supplementary learning material adaptively rendered using a prior data and complete all requirement to unlock another topic

Explore Computer Science

OVERALL PROGRESS **75%**

3 topics finished

RECENT TOPICS

- AI in Software Engineering Updated 0%
- AI in Software Engineering Updated today
- Topic threat-modeling 4%
- Understanding Requirements 19 weeks ago
- AI in Software Engineering 19 weeks ago

ACTIONS

Start New Topic **UNLOCKED**

Rated ✓

Practice Question

Review Concepts

AI in Software Engineering Updated

Learning Module - 6 min

0% completed

Progress requirements
Complete all items to unlock the next topic

✓

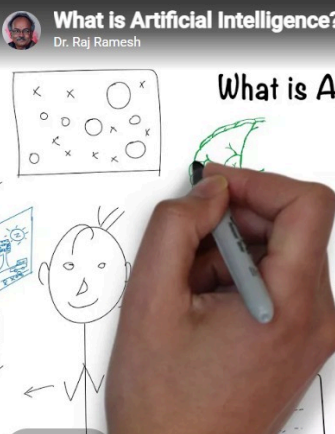
✓

✓

✓

1. Watch the video to the end
2. Complete the rating
3. Answer the practice question
4. Check the review concept

Unlock next topic



What is Artificial Intelligence? In 5 minutes.(Part 2 link in Description)

Dr. Raj Ramesh

What is AI - For People In a Hurry!

Computer Science

Image Processing

Symbolic learning

Machine Learning

Statistical Learning

Speech Recognition

NLP

Robotics

Computer Vision

Pat Rec

Computer Vision

My Learning Hub
Dept. of Computer Science

OVERALL PROGRESS 50%
2 topics finished

RECENT TOPICS

- Secure Design Patterns 9%
- Topic threat-modeling 38%
- Threat Modeling Fundamentals 38%
- AI in Software Engineering Upda... 2 weeks ago
- AI in Software Engineering 3 weeks ago

ACTIONS

Start New Topic UNLOCKED

Rated ✓

Threat Modeling Fundamentals

Learning Module · 22 min

0% completed

Progress requirements
Complete all items to unlock the next topic

✓

✓

✓

✓

1. Read the material to the end 2. Complete the rating 3. Answer the practice question 4. Check the review concept

Unlock next topic

Page 3 / 26 (2 of 26 read) -- Prev Next -- 100% + Progress: 8%

STRIDE and Associated Derivations

STRIDE is currently the most mature threat modeling method. Invented by Loren Kohnfelder and Praerit Garg in 1999 and adopted by Microsoft in 2002, STRIDE has evolved over time to include new threat-specific tables and the variants STRIDE-per-Element and STRIDE-per-Interaction [14, 20, 20].

65-B-09
University Of Alabama

TOPIC PROGRESS
0% Threat Modeling Fundamentals
Keep going!

Open FAQ

FAQ X Close

Search FAQ...

How do I logout?

How can I edit my profile?

How do I collapse or hide the Progress Tracker?

Privacy — what happens to my data?

Step 8: To unlock another topic, participant must read or watch the material to the end, rate the usefulness of the material, complete post test quiz and review AI generated concept

+ Transcript

How did you feel about this content?

I love this I like this I am neutral I dislike this I hate this

Please select the icon that best matches your feeling.

Why did you choose this rating?

Click a tag to append it, or type your own comment.

Too high-level Missing examples Pace too fast

Too high-level Missing examples

What is a significant role of AI in modern software development?

A Automating manual testing

B Creating graphics for user interface

C Writing project documentation

D Managing project timelines

Incorrect Choice

AI has a significant role in automating manual testing in modern software development, which helps to increase efficiency and reduce errors.

Finish Quiz Next Question

How does AI contribute to modern software development?

- A AI can generate code automatically
- B AI can predict software bugs
- C AI can optimize software performance

D All of the above

🎉 **Excellent! Correct Answer**

AI in software engineering can contribute in various ways such as automatic code generation, bug prediction, and performance optimization. Therefore, 'All of the above' is the correct answer.

Finish Quiz

Next Question

← **AI Concept Recap**
Accelerated personalized learning synthesis

AI in Software Engineering

📖 **TOPIC OBJECTIVES**

- ✓ Explore the role of AI in modern software development
- ✓ Understand AI-driven code generation
- ✓ Discuss ethical considerations in AI-powered tools

Regenerate Summary

🔖 Save to notes

● **KEY CONCEPTS**

- AI assists in automating repetitive tasks in software development
- AI-driven code generation uses machine learning to create code snippets
- Ethical considerations in AI tools include bias detection and transparency

● **APPLIED EXAMPLES**

- An AI tool that suggests code completions based on the context of the project

● **RECOMMENDED PRACTICE**

- Explore online courses on AI ethics in software development

Step 9: Student's dashboard to see their progress

**My Learning Hub**
Dept. of Computer Science

OVERALL PROGRESS **50%**

2 topics finished

RECENT TOPICS

- Secure Design Patterns
9%
- Topic threat-modeling
38%
- Threat Modeling Fundamentals
38%
- AI in Software Engineering Upda...
2 weeks ago
- AI in Software Engineering
3 weeks ago

ACTIONS

 **Start New Topic** **UNLOCKED**

AI in Software Engineering
19 weeks ago

ACTIONS

 Start New Topic

Rated ✓

 Practice Question

 Review Concepts

 Connect with Peers

Logout

 **65-B-09**
University Of Alabama

TOPIC PROGRESS

0%

Threat Modeling Fundamentals
Keep going!

Open FAQ

FAQ × Close

Search FAQ...

How do I logout? ▾

How can I edit my profile? ▾

How do I collapse or hide the Progress Tracker? ▾

Privacy — what happens to my data? ▾